Matthew Slipenchuk

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Experience

Murphy Cares — Philadelphia, PA
Lead Software Engineer (Contract) — *Oct. 2025 - Feb. 2026*

- Reduced client onboarding time 77% by building an admin platform enabling internal users to deploy and manage multi-tenant sites. (**Next.js, Terraform, AWS, GCP**)
- Developed and scaled a JWT-based authentication microservice supporting passcode and geolocation verification. (**FastAPI, Firebase, AWS ECS, OpenTelemetry, Grafana**)
- Implemented a Dataform pipeline for streamed app analytics, supporting incremental monthly external reporting dashboards. (**SQL, GCP BigQuery, Looker**)

Vitara Biomedical — Philadelphia, PA
Software Engineer — *Jan. 2021 - May 2025*

- Led 0 to 1 SDLC of FDA 510(k) clinical data platform via Agile IEC-62304:
  - Scaled 95-table lakehouse (Quality Tiers + STAR schema marts) to 1.56B records/day via Delta Live Tables, AutoLoader, clustering, and partitioning. (**Python, Databricks**)
  - Developed OLAP/OLTP API with adaptive downsampling, supporting 5ms-resolution queries over 30+ day windows. (**FastAPI, Databricks, Postgres**)
  - Optimized dashboard serving interactive +1M datapoint charts under 2s. (**Next.js**)
- Built services for Class III medical device (FDA IDE accepted), utilizing multi-node pub/sub routing for stateful alarms (250+), touch UI, and camera feed. (**C++17, Qt, MQTT, Linux**)
- Built an event-driven telemetry serialization microservice: processing 3.6M binary records per file into 30 CSVs with 200ms end-to-end latency. (**C++11/17, Qt5, MQTT, Azure Blob**)
- Automated FDA IDE data collection for 16 SKUs by building an EOL tracker platform capturing incremental metrics via Smartsheet Webhooks. (**Next.js, FastAPI, Postgres**)
- Deployed CI/CD triggered code generation agent to automate data schema parity across 4 codebases. Reduced data schema errors from 20% to 0% for releases post deployment.
- Integrated Claude Code into IEC-62304 compliant software development life cycle. Enabling code generation via TDD (exported TCs) and UAT gating via BDD (gherkin).
- Developed retrospective, off-policy Q-Network RL for heparin dosing; k-fold CV confirmed 1-SD improvement over clinicians, standardizing care during high lab turnover. (**Python**)
- Engineered regression-based blood oxygen system (R^2=0.97 vs Spectrum Medical Oximeter) with published design-around patent. (***C++, Python, R***) (US20250360251A1)

Children's Hospital of Philadelphia Research Institute — Philadelphia, PA
Software Developer (Contract) — *Jan. 2019 - Dec. 2020*

- Developed embedded devices, data streaming, and reporting systems enabling root-cause analysis and protocol optimization for artificial womb therapy. (**C++, Python, R**)

Skills

Languages: Python, TypeScript, JavaScript, C++11/17, Go, R, C, MATLAB, Ruby
Frameworks: FastAPI, Qt, NextJS, React, React Native, Vue, Ruby on Rails
Technologies: Azure (ADLSv2, ACR, AKS, IoT), AWS (S3, EC2, ECS, EKS, CloudFront), GCP (BigQuery, Dataform), Databricks, Postgres, DuckDB, Terraform, K8s, Docker, JWT, MQTT, Git

Education

Temple University — Philadelphia, PA
B.S. Computer Science — *Dec. 2020*